

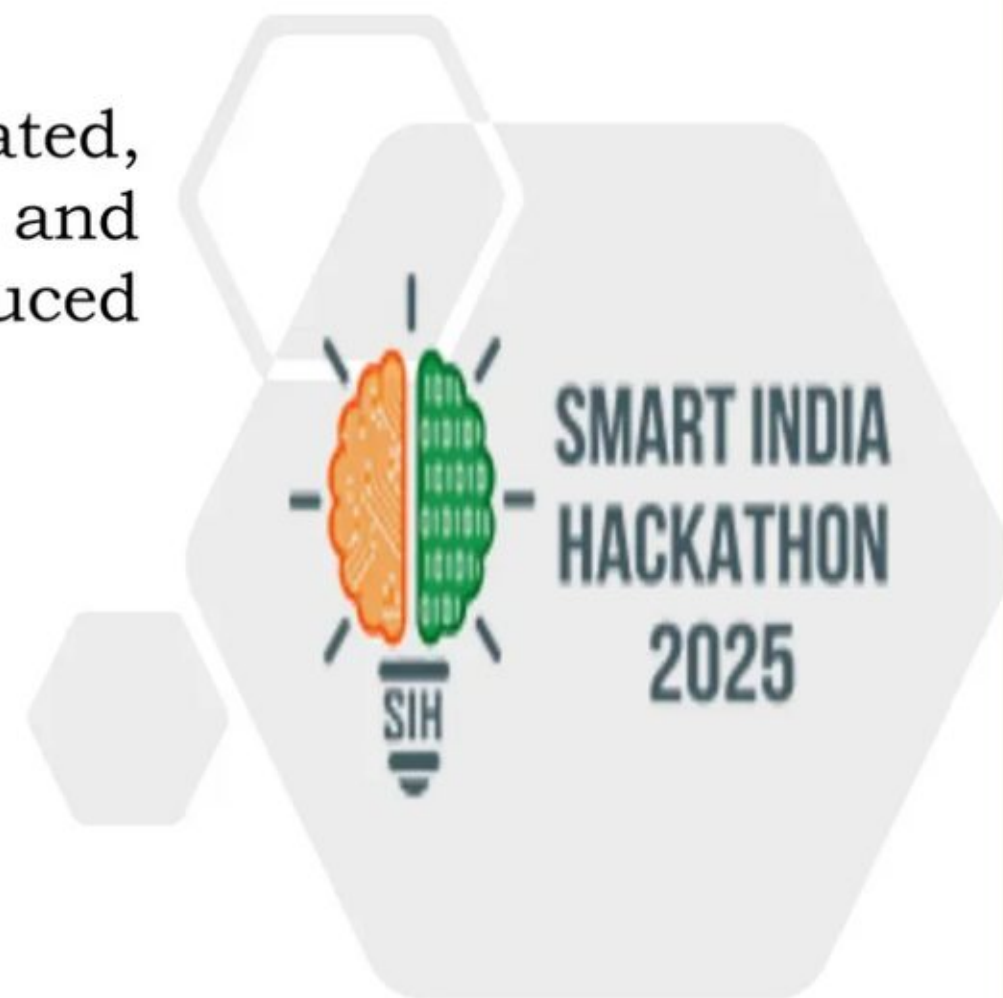


SMART INDIA HACKATHON 2025



Problem Statement ID –25099

- **Problem Statement** - Rice farmers ☐ lack an integrated, intelligent system to monitor crop health, soil condition, and energy usage — leading to wasted resources and reduced yield.
- **Theme-** Agriculture, Food Tech & Rural Development
- **PS Category-** Software/Hardware
- **Team ID –**
- **Team Name –** AI&DS Innovators



AGRI NEXA – AI + IoT FOR SMART AGRICULTURE

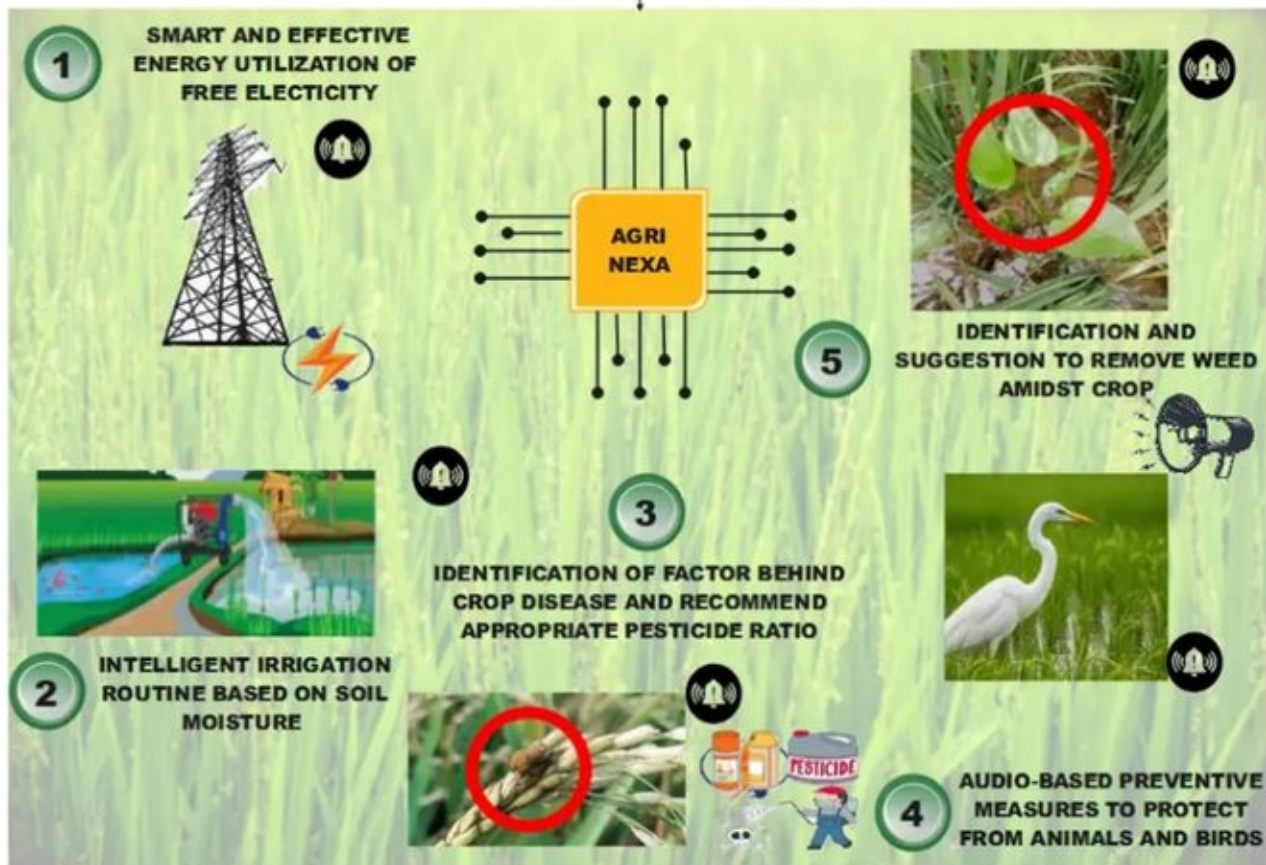
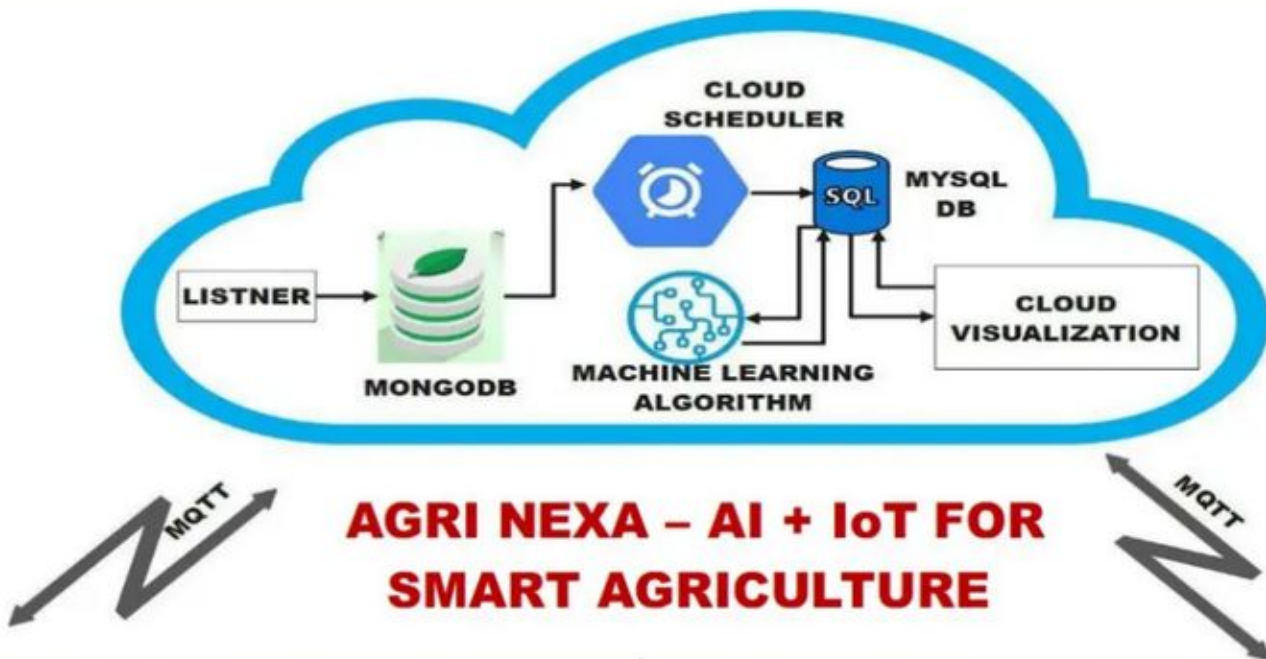


Turning Data into Smarter Yields

An **AI + IoT-powered smart farming system** tailored for rice cultivation that integrates sensors, machine learning, and cloud analytics to optimize water, energy, and crop health management.

- ☐ **Soil & Crop Monitoring** → Sensors track moisture, nutrients, and plant health in real time.
- ☐ **Intelligent Irrigation** → Automated water supply based on soil condition and crop stage.
- ☐ **Pest & Disease Detection** → AI analyzes field/sensor data to recommend pesticide use.
- ⚡ **Energy Optimization** → Leverages free electricity hours for irrigation and farm operations.
- 🐦 **Bird & Animal Prevention** → Smart audio deterrents protect rice fields.
- ☐ **Weed Detection** → AI models identify weeds and suggest removal strategies.





☆ Uniqueness of AGRI-NEXA

1. Rice-farm Specific Focus - Tailored to the needs of **paddy fields** where water, pests, and energy are critical.

2. Energy-Aware Farming - First solution to integrate **free electricity slot tracking** with smart irrigation.

3. AI + IoT Fusion - Combines **sensor data + AI models** (soil, pest, weed, disease, electricity) into a **single decision engine**.

4. End-to-End Automation - From **soil sensing** → **irrigation control** → **pest deterrent** → **farmer dashboard**, all in one ecosystem.

5. Cost-Effective & Scalable - Uses **low-cost sensors** (soil, current, servo motors) + cloud AI, making it **affordable for small farmers**.

6. Preventive Agriculture - Focuses on **early detection and prevention** (pests, weeds, diseases) instead of just reactive measures.

TECHNICAL APPROACH

AGRI NEXA - AI Vs. IoT



AI Approach

- **Irrigation Forecasting** → LSTM predicts soil moisture & schedules water usage
- **Disease Detection** → KNN (leaf images) identifies rice diseases early
- **Weed Detection** → YOLO detects unwanted weeds
- **Pest Risk Prediction** → XGBoost recommends spray timing & dosage
- **Bird/Animal Prevention** → CNN triggers smart sound deterrents
- **Energy Optimization** → AI aligns irrigation with free electricity hours
- **Decision Fusion** → AI engine merges soil, image & energy data

IoT Architecture

- **Soil Sensors** (moisture, temp, EC) send live data
- **Camera Modules** capture leaf/field images
- **Weed Alerts** sent via mobile dashboard
- **Pest Sensors / Image Traps** feed risk data
- **Audio Devices** (speaker + servo motor) for deterrence
- **Smart Relays + Current Sensors** monitor pumps & energy use
- **Cloud + MQTT Broker** connect sensors → DB → Dashboard

FEASIBILITY AND VIABILITY



□ Feasibility

- Affordable IoT sensors (soil, current, servo)
- Proven AI/ML techniques in farming
- Scalable cloud integration (MQTT + DB)
- Simple farmer-friendly dashboard

💰 Viability

- Saves **water (30% approx)** & **energy cost**
- Boosts rice yield by **15–20%**
- Easy adoption by local farmers
- Supports sustainable agriculture goals



IMPACT AND BENEFITS

- **Water Conservation** → Up to **30% savings** through intelligent irrigation.
- **Energy Efficiency** → Optimized use of **free electricity slots** lowers costs.
- **Higher Yield** → Improved crop health boosts rice yield by **15–20%**.
- **Reduced Losses** → Early pest/disease/weed detection prevents damage.
- **Data-Driven Farming** → Real-time insights for smarter decisions.
- **Sustainability** → Supports eco-friendly and resource-efficient agriculture.



RESEARCH AND REFERENCES

- https://tnlasdigital.tn.gov.in/jspui/free_electricity.jsp
- <https://www.newindianexpress.com/states/tamil-nadu/2025/Apr/26/farmers-upset-with-tne-b-circular-barring-free-electricity-to-solar-power-user>
- <https://www.academicx.org/SAE/2026/>
- https://www.iconf.com/conference/SAE2026_8645
- <https://www.iconf.com/conference/icsea2022>
- <http://www.icsea.org/>
- <https://elibrary.asabe.org/abstract.asp?aid=54809>
- <https://link.springer.com/article/10.1007/s43069-025-00434-z>
- https://scholar.google.com/scholar?hl=en&as_sdt=0%2C5&q=AI%2FML%2BBIOT%2Bsmart+farming+irrigation+elseviers&oq=
- <https://www.jestr.org/downloads/Volume18Issue3/fulltext141832025.pdf>



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THANK YOU